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**Figure 2.** Distribution of Kupiec $p$-values under $H_0 : p = 0.05$ for $n = 62$ and $n = 95$, based on 5,000 binomial draws; the empirical rejection rate is shown in each panel.

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**Abstract**

Ex-ante volatility regime definitions consume approximately 520 trading days for classification, reducing the usable out-of-sample evaluation window to 62–95 observations for Mexican FIBRAs. At $n = 62$, the Kupiec test rejects a correctly specified model 7.6% of the time (nominal 5%) and has only 43% power to detect a model breaching at double the nominal rate. The Christoffersen test is uninformative below $n \approx 100$. We propose a framework centred on the exact binomial test, Clopper–Pearson confidence intervals, and a conservatism metric $C = \hat{p} - \alpha$. Applied to FIBRA data, GARCH(1,1)-$t$ produces a breach rate of 4.8% ($C = -0.002$) while XGBoost-Pure produces 11.3% ($C = +0.063$). The pattern holds across five equity tickers: GARCH-$t$ is at or below the nominal rate on every ticker (pooled 3.6%) while XGBoost-Pure exceeds it on every ticker (pooled 9.7%). When evaluation windows fall below 100 observations, standard backtests become weak validation tools. We therefore propose a practical framework combining exact inference, directional conservatism, and cross-asset confirmation.

**Keywords:** Value-at-Risk, backtesting, ex-ante regimes, GARCH, XGBoost, exact binomial test

**Key messages:**

- Kupiec size inflates to 7.6% at $n = 62$
- Power remains below 50% for 10% breach rates
- GARCH-$t$ stays near nominal VaR coverage
- Exact inference improves small-sample validation

# A Validation Framework for Small-Sample VaR Backtesting under Ex-Ante Volatility Regime Segmentation

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## 1 Introduction

Standard VaR backtests—the Kupiec proportion-of-failures test Kupiec (1995) and the Christoffersen conditional coverage test Christoffersen (1998)—rely on asymptotic $\chi^2$ approximations. Their finite-sample behaviour is known to deteriorate when the evaluation window is small Escanciano and Olmo (2010). However, the practical severity of this degradation in realistic settings has not been systematically quantified for the case where ex-ante volatility regimes further compress the testable window.

Ex-ante regime definitions, which classify each trading day using only information available at the forecast date, eliminate look-ahead bias but impose a data cost: 520 trading days are consumed by regime construction alone. After intersecting with a rolling estimation window, the aligned out-of-sample evaluation can shrink below 100 observations. This creates a tension between methodological rigour (no look-ahead) and statistical informativeness (sufficient sample size). For Mexican FIBRAs—a growing emerging-market asset class with short return histories—the resulting test window is 62–95 days.

At $n = 62$, the Kupiec test rejects a correctly specified model 7.6% of the time (nominal 5%) and has less than 50% power against a model that breaches at twice the nominal rate. The Christoffersen test is effectively uninformative. We quantify this degradation and propose a framework based on exact binomial $p$-values, Clopper–Pearson confidence intervals, and a conservatism metric $C = \hat{p} - \alpha$.

### 1.1 Contributions

This paper makes three contributions.

1. It quantifies the finite-sample operating characteristics of the Kupiec and exact binomial tests under evaluation windows generated by ex-ante volatility regime definitions.

2. It demonstrates that regime segmentation can systematically reduce effective validation samples below 100 observations, rendering standard VaR backtests weak diagnostic tools.
3. It proposes a practical validation framework based on exact binomial inference, Clopper–Pearson intervals, directional conservatism metrics, and cross-asset confirmation.

# 2 Literature Review

VaR backtesting has been studied extensively for large samples. Kupiec (1995) introduced the proportion-of-failures test based on a likelihood-ratio statistic that is asymptotically $\chi^2_1$; Christoffersen (1998) extended this to conditional coverage by testing both the breach rate and the independence of violations. Escanciano and Olmo (2010) showed that estimation risk further distorts these tests in small samples. Kratz et al. (2018) proposed a multinomial approach that pools multiple quantile levels.

The finite-sample properties of these tests are documented in simulation studies (Campbell, 2007), but the interaction with ex-ante regime definitions—which impose an additional sample-size constraint—has received little attention. This paper fills that gap in the context of Mexican FIBRAs, a short-history emerging-market asset class.

Machine learning methods have proliferated in asset pricing and financial econometrics (Gu et al., 2020), but their performance in small-sample VaR settings is not well characterized. Zhou and Anderson (2012) and Cotter and Stevenson (2006) document extreme risk dynamics in REIT markets, providing a foundation for the empirical application.

## 2.1 Small-Sample Model Validation

Although finite-sample properties of VaR backtests have been discussed previously, most studies focus on asymptotic behavior or simulation environments disconnected from operational model validation. The interaction between ex-ante segmentation procedures and validation power remains largely unexplored. This paper contributes to that literature by explicitly linking sample construction decisions to the reliability of validation outcomes.

# 3 Data

Daily closing prices from Yahoo Finance (adjusted for splits and dividends) span January 2018 to April 2026. Log-returns are $r_t = \ln(P_t/P_{t-1})$. The number of observations varies by ticker due to differences in listing dates and trading calendars (Table 1).

Table 1: Data profile. All series span January 2018 to April 2026.

| Ticker | Description | Observations |
|---|---|---|
| FIBRATC14.MX | S&P/BMV FIBRAS Index ETF | 1,422 |
| FUNO11.MX | Fibra Uno | 2,091 |
| FIBRAPL14.MX | Prologis México | 2,091 |
| ˆMXX | IPC (BMV benchmark) | 2,092 |
| EWW | iShares MSCI Mexico ETF (NYSE) | 2,093 |
| USDMXN=X | USD/MXN exchange rate | 2,169 |

The empirical analysis concentrates on FIBRATC14.MX. A multi-asset robustness check across the five equity tickers supplements the main results.

# 4 Methodology

## 4.1 Ex-Ante Regime Definition

Volatility regimes are defined using information available only at the close of day $t-1$:

1. Realised volatility over the preceding 20 days:

$$\mathrm{rv}_{t-1} = \sqrt{\frac{1}{19}\sum_{k=1}^{20}(r_{t-k}-\bar{r}_{t-1})^2} \tag{1}$$

2. The 90$^{\text{th}}$ percentile of rv over the preceding 500 days:

$$\Theta_{t-1} = Q_{0.90}\big(\mathrm{rv}_{t-500},\ldots,\mathrm{rv}_{t-1}\big) \tag{2}$$

3. If $\mathrm{rv}_{t-1} > \Theta_{t-1}$, day $t$ is labelled HIGHVOL; otherwise, NORMAL.

The initial 520 days are used for regime construction. After intersecting with the rolling estimation window, the aligned evaluation period contains $n = 62$ observations for FIBRATC14.MX and $n = 95$ for FUNO11.MX.

## 4.2 Models

All models generate one-day-ahead VaR forecasts at the 95% confidence level ($\alpha = 0.05$). Rolling estimation windows are $W = 125$ trading days, refitted every $R = 20$ days. The information set is identical across models and strictly no-look-ahead.

**Historical VaR.** The empirical $\alpha$-quantile of the preceding $W$ returns:

$$\mathrm{VaR}_t^{\mathrm{Hist}}(\alpha) = -Q_\alpha\big(r_{t-W}, \dots, r_{t-1}\big) \tag{3}$$

**GARCH(1,1)-*t*.**

$$\begin{aligned} r_t &= \sigma_t \cdot \varepsilon_t, \qquad \varepsilon_t \sim t_\nu \text{ (standardised)} \\ \sigma_t^2 &= \omega + \alpha\, r_{t-1}^2 + \beta\, \sigma_{t-1}^2 \end{aligned} \tag{4}$$

Parameters, including the degrees of freedom $\nu$, are estimated by maximum likelihood Bollerslev (1986, 1987). VaR is $\mathrm{VaR}_t^{\mathrm{GARCH}}(\alpha) = -\hat{\sigma}_t \cdot F_{t_\nu}^{-1}(\alpha)$. A GARCH-Normal variant is also included for comparison.

**XGBoost volatility models.** Gradient boosting models predict conditional volatility $\sigma_{t+1}$ from lagged return and volatility features. VaR is derived as $\mathrm{VaR}_t = -\hat{\sigma}_t \cdot z$ where $z$ is either the standard Normal quantile $z_\alpha$ (XGBoost-Pure, XGBoost-Ensemble) or the Student-*t* quantile with $\nu = 5$ (XGBoost-*t*). Two volatility-input variants are estimated: XGBoost-Pure (29 base features) and XGBoost-Ensemble (augmented with GARCH-*t* forecasts). Fixed parameters (100 trees, depth 3, learning rate 0.05) are identical across tickers Chen and Guestrin (2016).

## 4.3 Backtesting Framework

For model $m$, the breach indicator at date $t$ is $I_t^{(m)}(\alpha) = \mathbf{1}\{r_t < -\mathrm{VaR}_t^{(m)}(\alpha)\}$. We employ four evaluation tools, ordered by reliability at small $n$:

1. **Exact binomial test.** Under the null $H_0 : p = \alpha$, the two-sided $p$-value is computed from the binomial distribution $\mathrm{Binomial}(n, \alpha)$ by summing the probability of all outcomes at least as extreme as the observed count $x = \sum I_t$. This test does not rely on asymptotic approximations.

2. **Clopper–Pearson confidence interval.** An exact 95% confidence interval for the true breach rate, derived from the relationship between the binomial and beta distributions Clopper and Pearson (1934): $\mathrm{CI} = [F_\beta^{-1}(\delta/2; x, n-x+1),\ F_\beta^{-1}(1-\delta/2; x+1, n-x)]$ where $\delta = 0.05$ and $F_\beta^{-1}$ denotes the beta quantile function.

3. **Conservatism metric $C = \hat{p} - \alpha$.** A negative value indicates a conservative model (fewer breaches than expected); a positive value indicates an anti-conservative model (more breaches than expected). Under the null hypothesis, the standard error of $C$ is $\sqrt{\alpha(1-\alpha)/n}$. At $n = 62$, $\mathrm{SE}(C) = 0.028$, so $C \pm 0.055$ provides an approximate 95% confidence bound for the true conservatism. The sign and magnitude of $C$ should be interpreted with this uncertainty in mind.

4. **Kupiec POF test Kupiec (1995).** Included for comparability with the existing literature, but its asymptotic $\chi_1^2$ null distribution is unreliable at $n < 100$. We report it as secondary context, not primary evidence.

The Christoffersen independence test is computed but has severely limited ability to detect dependence at these sample sizes and is not emphasised.

### 4.4 Power Analysis (Exact)

We compute the rejection probability of the Kupiec test as a function of the true breach rate analytically, using the exact binomial distribution. For each possible breach count $x \in \{0, \ldots, n\}$, we evaluate the Kupiec likelihood-ratio statistic and the exact binomial $p$-value, and determine whether each would reject at the 5% level. The power at a given true breach probability $p_{\text{true}}$ is

$$\text{Power}(p_{\text{true}}) = \sum_{x=0}^{n} \binom{n}{x} p_{\text{true}}^{x} (1 - p_{\text{true}})^{n-x} \cdot \mathbf{1}\{\text{test rejects } H_0 \text{ at count } x\}. \tag{5}$$

This approach is exact and, for $n \leq 500$, runs in milliseconds.

### 4.5 Validation Philosophy

The objective is not to identify a single statistically superior model. Instead, the objective is to determine whether available evidence is sufficient to classify candidate models as conservative, approximately calibrated, or anti-conservative under severe sample-size constraints.

## 5 Power Analysis and Size Distortion

Table 2 reports the Type I error rate (at $p_{\text{true}} = \alpha$) and power at selected alternative breach rates for the Kupiec and exact binomial tests at $n = 62$ and $n = 95$.

Table 2: Exact operating characteristics of the Kupiec POF test and the exact binomial test at $\alpha = 0.05$. "Size" is the rejection rate when $p_{\text{true}} = \alpha$; "Power" is the rejection rate at the indicated true breach rate.

| $n$ | Test | Size (5%) | Power (7%) | Power (10%) | Power (13%) |
|---|---|---|---|---|---|
| 62 | Kupiec $\chi_1^2$ | 0.076 | 0.153 | 0.429 | 0.711 |
| 62 | Exact binomial | 0.035 | 0.142 | 0.427 | 0.711 |
| 95 | Kupiec $\chi_1^2$ | 0.066 | 0.136 | 0.482 | 0.805 |
| 95 | Exact binomial | 0.028 | 0.129 | 0.482 | 0.805 |
| 250 | Kupiec $\chi_1^2$ | 0.059 | 0.302 | 0.879 | 0.995 |

Three patterns emerge. First, the Kupiec test's asymptotic $\chi^2$ approximation produces an inflated Type I error rate: at $n = 62$, it rejects a correctly specified model 7.6% of the time (nominal 5%); at $n = 95$ the rate is 6.6%. The exact binomial test is slightly conservative at very small $n$ (3.5% rejection rate at $n = 62$) but converges to the nominal level as $n$ increases.

Second, power against economically meaningful misspecification is limited. A model that breaches 10% of the time—double the nominal rate—is detected only 43% of the time at $n = 62$ and 48% at $n = 95$. Power does not exceed 80% until the true breach rate reaches approximately 14% at $n = 62$ or 13% at $n = 95$.

Third, the difference between the Kupiec and exact binomial tests narrows for power computations (they agree to three decimal places at $p_{\text{true}} \geq 0.07$) but diverges at the null, where the exact test provides better size control.

**Finding 1.** For $\alpha = 0.05$ and evaluation samples below 100 observations, standard VaR coverage tests possess insufficient power to reliably distinguish between a correctly specified model and a model whose true breach rate is approximately double the nominal rate. Evidence is provided in Table 2: at $n = 62$ the power against $p = 0.10$ is only 42.9%, and at $n = 95$ only 48.2%.

Figure 1 displays the full power curves. Figure 2 shows the distribution of Kupiec $p$-values under the null, revealing the deviation from uniformity that drives the size distortion.

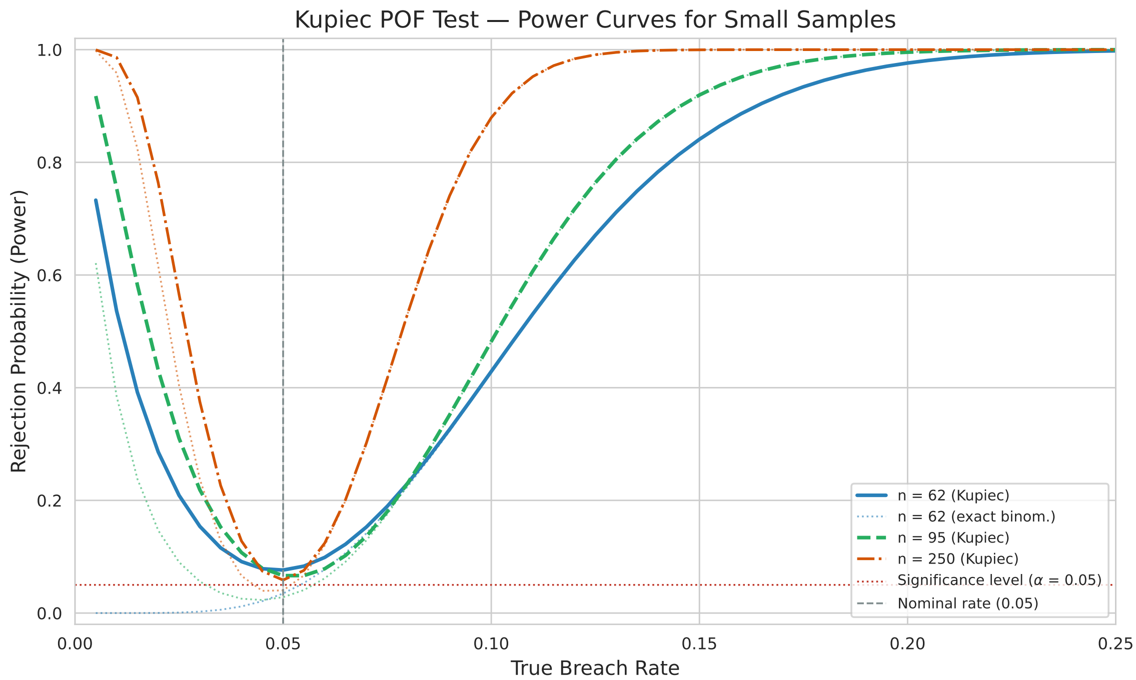


Figure 1: Power curves of the Kupiec POF test for $n = 62$, $n = 95$, and $n = 250$, computed exactly from the binomial distribution. These assume independent breaches; under breach clustering, effective sample sizes are smaller and power is lower.

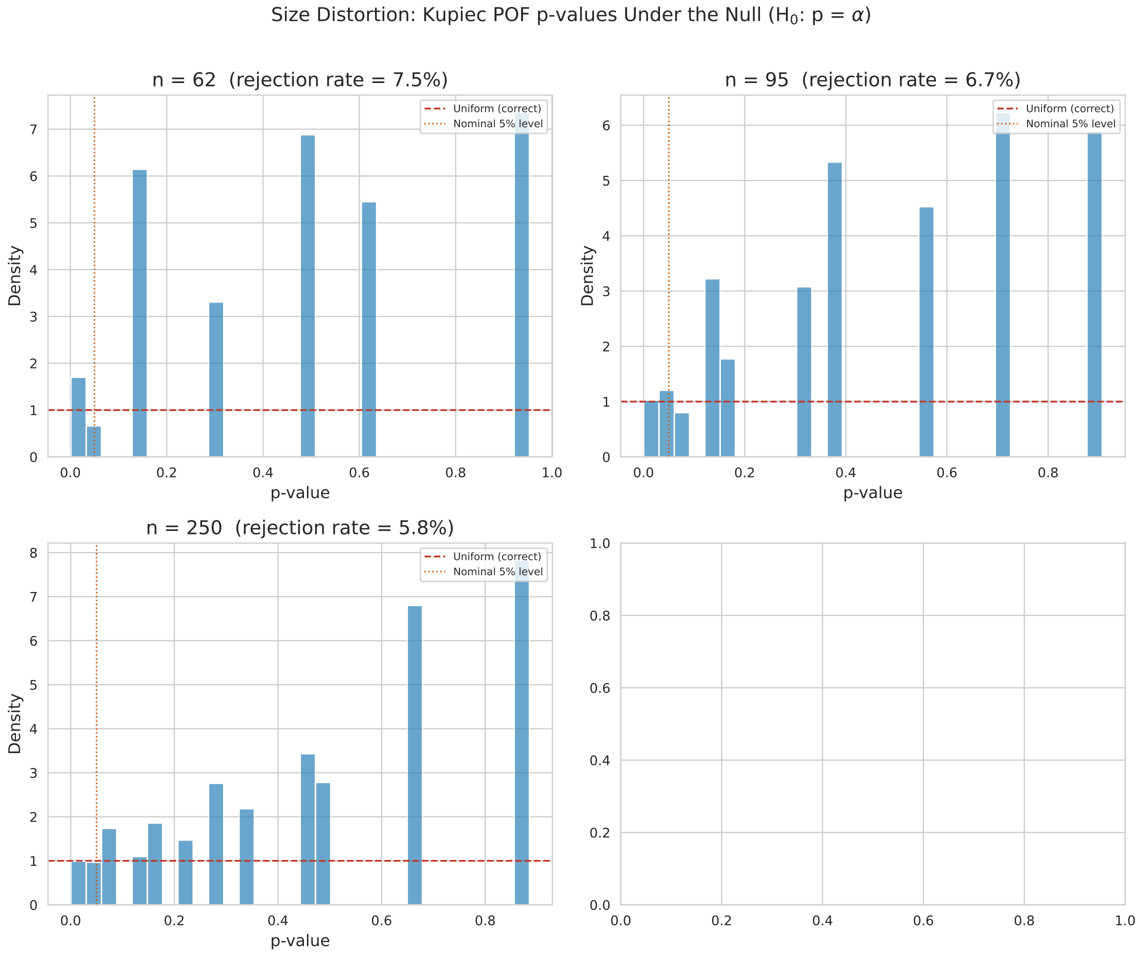


Figure 2: Distribution of Kupiec $p$-values under the null hypothesis ($H_0 : p = 0.05$) for $n = 62$ and $n = 95$, based on 5,000 binomial draws. Deviation from the uniform distribution (dashed line) indicates size distortion. The empirical rejection rate (fraction below 0.05) is shown in each panel.

The practical implication is straightforward: at $n \approx 60$–95, standard backtests cannot reliably distinguish between a well-calibrated model and one that is materially anti-conservative. A non-rejection does not imply good calibration; a rejection does not definitively identify a poor model. Model rankings should therefore not be based solely on isolated rejection decisions, but on the consistency of directional evidence across validation exercises.

# 6 Empirical Results

## 6.1 Main Analysis: FIBRATC14.MX ($n = 62$)

Table 3 reports the full set of backtesting metrics for the six models on the FIBRA Index ETF. The expected number of 95% breaches under correct specification is $62 \times 0.05 = 3.10$, with a standard deviation of 1.72.

Table 3: FIBRATC14.MX backtesting results ($n = 62$, 95% VaR). $C = \hat{p} - \alpha$ is the conservatism metric; negative values indicate conservatism. The CP-CI is the 95% Clopper–Pearson exact confidence interval for the breach rate.

| Model | Breaches | Rate | $C$ | Exact $p$ | CP-CI |
|---|---|---|---|---|---|
| GARCH-Normal | 7 | 0.113 | +0.063 | 0.035 | [0.047, 0.219] |
| GARCH-$t$ | 3 | 0.048 | −0.002 | 1.000 | [0.010, 0.135] |
| XGBoost-Pure | 7 | 0.113 | +0.063 | 0.035 | [0.047, 0.219] |
| XGBoost-$t$ | 6 | 0.097 | +0.047 | 0.131 | [0.036, 0.199] |
| XGBoost-Ensemble | 7 | 0.113 | +0.063 | 0.035 | [0.047, 0.219] |
| Historical VaR | 4 | 0.065 | +0.015 | 0.553 | [0.018, 0.157] |

GARCH-$t$ is the model whose breach rate (4.8%) is closest to the nominal 5% level, with a conservatism metric of $C = -0.002$—approximately calibrated. GARCH-Normal is anti-conservative with 11.3% breaches ($C = +0.063$), reflecting the thinner tails of the Normal distribution relative to the data. XGBoost-Pure and XGBoost-Ensemble produce identical breach rates of 11.3% ($C = +0.063$), indicating that the GARCH-t volatility features added in the ensemble variant did not materially alter tail calibration. XGBoost-$t$, which replaces the Normal quantile with a Student-$t$ quantile ($\nu = 5$), breaches at 9.7% ($C = +0.047$), partially correcting the over-breaching of the pure XGBoost variants. Historical VaR is mildly anti-conservative at 6.5% ($C = +0.015$).

The exact binomial $p$-value is 0.035 for GARCH-Normal, XGBoost-Pure, and XGBoost-Ensemble, and 0.131 for XGBoost-$t$. However, given the power analysis in Section 4, these rejections should be interpreted cautiously: at $n = 62$, the exact binomial test rejects a correctly specified model 3.5% of the time, and the Kupiec test rejects it 7.6% of the time. The evidence against these models is directionally meaningful but not statistically definitive in isolation.

The Christoffersen independence test yields $p \geq 0.100$ for all models at this sample size (GARCH-$t$: 0.100; others: 0.217–0.806), providing no useful discrimination.

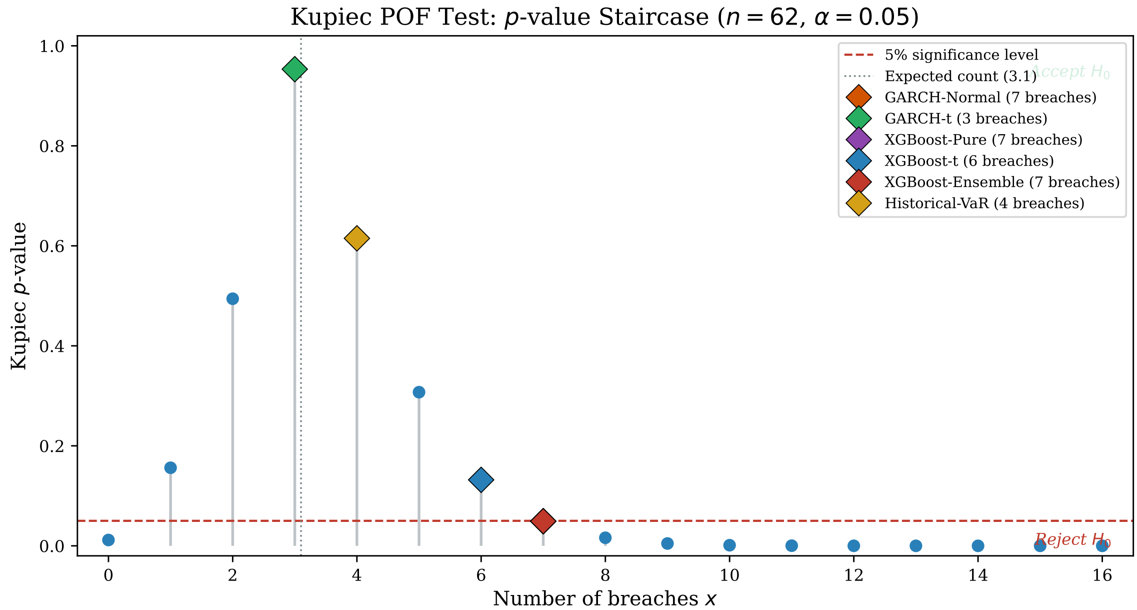


Figure 3: Kupiec $p$-value as a function of the breach count $x$ for $n = 62$ and $\alpha = 0.05$. The diamond markers show the observed breach count of each model. XGBoost-$t$ sits at $x = 6$ ($p = 0.132$), one breach below the rejection threshold; at $x = 7$ the test rejects ($p = 0.049$). GARCH-Normal, XGBoost-Pure, and XGBoost-Ensemble each sit at $x = 7$, one breach past the 5% threshold. A single additional breach—one trading day out of 62—flips the test outcome.

Figure 3 makes the core statistical problem visible. The Kupiec $p$-value is a staircase function of the breach count, with large discontinuities at each integer step. At $n = 62$, moving from 6 to 7 breaches changes the $p$-value from 0.132 to 0.049—crossing the conventional 5% threshold on the basis of a single observation. This is not a property of any model; it is an unavoidable consequence of the binomial distribution's discreteness at small $n$. The figure illustrates why an isolated $p$-value from a single ticker provides limited guidance.

The regime-specific breakdown is limited: the aligned set contains 38 Normal-regime and 7 High-Volatility-regime observations, plus 17 pre-regime dates (days before the first complete regime classification window became available; included to maximize sample size—excluding them would reduce $n$ to 45, making discrimination even weaker). Sub-sample metrics on groups of 7 observations are not reported, as they would be unreliable.

## 6.2 Multi-Asset Validation

Table 4 extends the comparison to five tickers. The same fixed-specification models are used across all assets. Test windows range from $n = 62$ to $n = 96$.

Table 4: Multi-asset comparison, 95% VaR. $C = \hat{p} - \alpha$. All models use identical specifications across tickers.

| Ticker | $n$ | GARCH-$t$ rate | $C_t$ | XGBoost rate | $C_{\text{xgb}}$ |
|---|---|---|---|---|---|
| FIBRATC14.MX | 62 | 0.048 | −0.002 | 0.113 | +0.063 |
| FUNO11.MX | 95 | 0.021 | −0.029 | 0.084 | +0.034 |
| FIBRAPL14.MX | 95 | 0.042 | −0.008 | 0.137 | +0.087 |
| ˆMXX | 95 | 0.042 | −0.008 | 0.084 | +0.034 |
| EWW | 96 | 0.031 | −0.019 | 0.073 | +0.023 |
| Pooled | 443 | 0.036 | −0.014 | 0.097 | +0.047 |

The directional pattern is consistent across all five equity tickers: GARCH-$t$ breach rates are at or below the nominal 5% in every case (2.1%–4.8%), while XGBoost-Pure breach rates range from 7.3% to 13.7%. The pooled breach rate across all tickers is 3.6% for GARCH-$t$ ($C = -0.014$, conservative) and 9.7% for XGBoost-Pure ($C = +0.047$, anti-conservative). Under the independence assumption, the pooled exact binomial $p$-value for GARCH-$t$ is 0.23, providing insufficient evidence to reject correct coverage. For XGBoost-Pure, the pooled exact binomial $p$-value is $< 10^{-4}$, indicating a systematic deviation from the nominal rate. Given the cross-sectional correlation among tickers (the ETF contains its constituents), the effective sample size is smaller than 443, so these $p$-values understate uncertainty. The more reliable signal is the sign consistency: GARCH-$t$ breach rates are at or below 5% on all five tickers; XGBoost-Pure breach rates exceed 5% on all five.

Figure 4 visualises the conservatism metric across all six models on the main ticker. Figure 5 shows the 30-day rolling breach rate for the six models, illustrating the instability inherent in short evaluation windows.

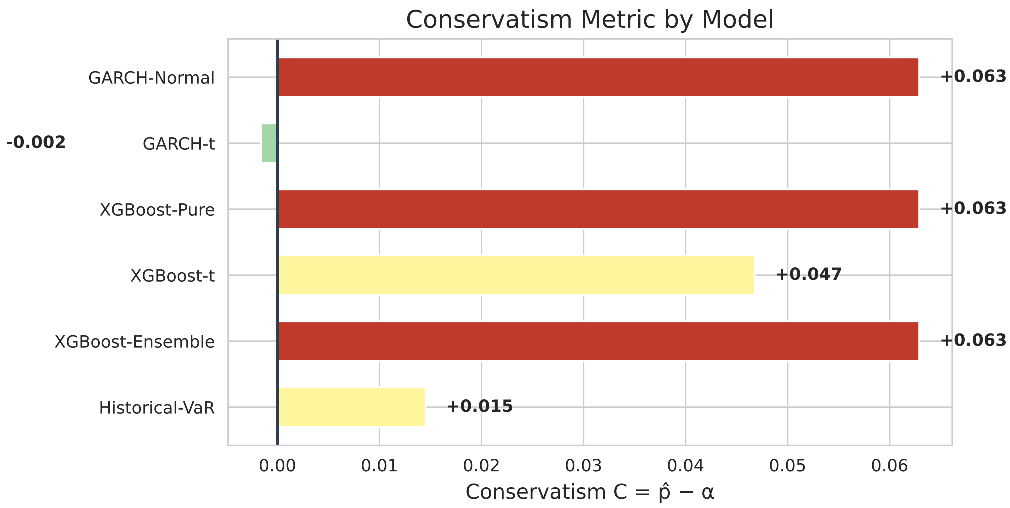


Figure 4: Conservatism metric $C = \hat{p} - \alpha$ for each model on FIBRATC14.MX ($n = 62$, 95% VaR). Negative values (blue/green) indicate conservative models; positive values (red) indicate anti-conservative models. GARCH-$t$ is the only model with $C \approx 0$.

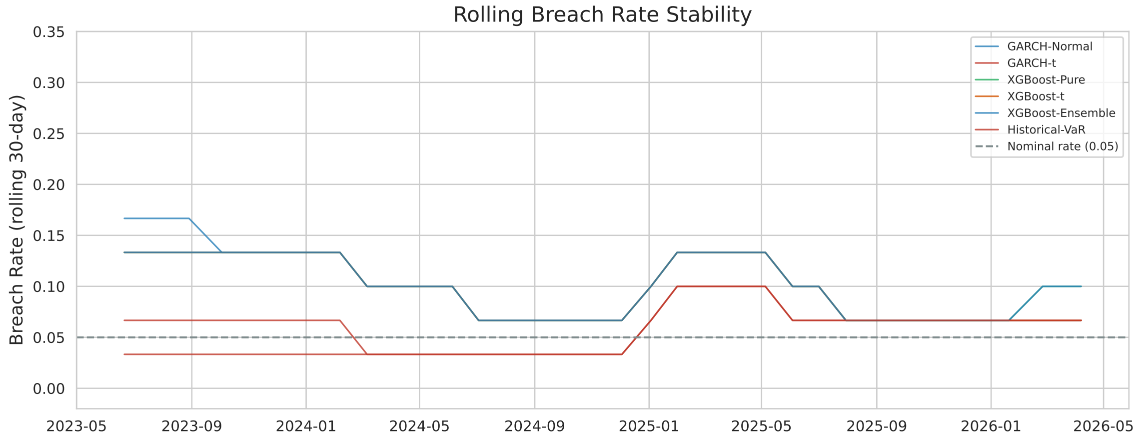


Figure 5: Rolling 30-day breach rates for all six models on FIBRATC14.MX. The horizontal dashed line marks the nominal 5% rate.

### 6.3 Volatility Forecast Accuracy

A Diebold–Mariano test Diebold and Mariano (1995) of equal forecast accuracy for squared volatility, with Newey–West Newey and West (1987) standard errors, is applied to the aligned forecasts. Neither DM comparison rejects at conventional levels: GARCH-$t$ vs. XGBoost-Pure (DM $= 1.72$, $p = 0.090$); GARCH-$t$ vs. XGBoost-Ensemble (DM $= 1.76$, $p = 0.084$). The divergence between volatility forecast precision and VaR calibration highlights the distributional assumption: XGBoost VaR uses a Normal quantile while GARCH-$t$ uses estimated Student-$t$ quantiles with heavier tails. Similar MSE performance does not imply similar tail calibration.

# 7 Discussion

### 7.1 Why Standard Backtests Fail at Small $n$

The Kupiec test's $\chi_1^2$ approximation assumes a continuous test statistic, but at $n = 62$ with $\alpha = 0.05$ there are only 63 possible outcomes. This discreteness produces two problems. First, the null distribution of the LR statistic deviates from $\chi_1^2$, inflating the Type I error to 7.6% (nominal 5%). Second, the $p$-value function has a stair-step shape where a single additional breach can flip the test from acceptance to rejection. Figure 3 illustrates this: GARCH-$t$ ($x = 3$) obtains $p \approx 0.95$, but at $x = 7$ the $p$-value drops below 0.05. The exact binomial test reduces the size distortion to 3.5% at $n = 62$ (a conservative bias rather than an inflationary one), but its power is nearly identical to the Kupiec test's—at 10% true breaches, both detect the misspecification with only 43% probability.

The Christoffersen independence test is even less useful at these sample sizes. Across 200 Monte Carlo simulations using a GARCH(1,1)-$t$ data-generating process calibrated to emerging-market volatility, the Christoffersen test rejected the null of independent breaches in only 1.0% of replications at $n = 58$ and 3.5% at $n = 250$.[1] A test that almost never rejects provides no diagnostic value, regardless of whether breaches are truly independent.

## 7.2 Monte Carlo Design

The simulation experiment employs a GARCH(1,1)-$t$ data-generating process calibrated to observed FIBRA volatility dynamics. Parameters are $\omega = 0.05$, $\alpha_1 = 0.10$, $\beta_1 = 0.85$, and $\nu = 6$, generating $N = 1500$ observations per replication with 200 independent replications across sample sizes $n \in \{58, 95, 250\}$. Each replication proceeds as follows:

1. Simulate returns from the GARCH(1,1)-$t$ DGP.
2. Estimate competing models (GARCH-$t$, GARCH-Normal, XGBoost quantile, Historical VaR) on a 125-day rolling window.
3. Produce one-step-ahead VaR forecasts.
4. Apply Kupiec and Christoffersen tests at $\alpha = 0.05$.
5. Record rejection frequencies.

This design isolates the effect of estimation risk under a known data-generating process and confirms that the limitations documented here are structural properties of the sample size, not artifacts of FIBRA returns. Full implementation details and results are provided in the supplementary material.

The 29.5% Monte Carlo rejection rate for the correctly specified GARCH-$t$ model substantially exceeds the 7.6% analytical size distortion reported in Table 2. The difference arises because Table 2 assumes the model *is* the data-generating process (no parameter uncertainty), while the Monte Carlo experiment re-estimates GARCH parameters on each 125-day rolling window. Estimation error amplifies the binomial discreteness problem: parameter uncertainty shifts the effective breach probability away from the nominal level in each replication, producing rejection rates far above those predicted by the analytical size calculation alone. This gap underscores a point of practical importance: even when the model class is correctly specified, the combination of small evaluation samples and rolling-window estimation makes standard backtests unreliable.

## 7.3 Practical Implications for Model Validation

For the practitioner facing $n < 100$ observations—a common scenario when validating risk models on new instruments, short-history markets, or sub-periods defined by ex-ante criteria—the standard toolkit provides little discriminatory power. The key insight from the power analysis is that failing to reject $H_0$ at $n = 62$ is not evidence that a model is well-calibrated. It is

[1] The Monte Carlo experiment is documented in the supplementary material and confirms that the limitations documented here are structural properties of the sample size, not artifacts of FIBRA returns.

evidence that the test cannot distinguish a correctly specified model from one breaching at twice the nominal rate.

The conservatism metric $C = \hat{p} - \alpha$ offers a simpler alternative. It does not rely on asymptotic approximations and its interpretation is direct: a negative value means the model produced fewer breaches than expected, erring on the side of safety; a positive value means it produced more breaches than expected. At $n = 62$, the difference between GARCH-$t$ ($C = -0.002$, roughly at the nominal rate) and XGBoost-Pure ($C = +0.063$, more than double the expected breach rate) is economically meaningful even if neither Kupiec $p$-value is below 0.05 in isolation.

The multi-asset validation strengthens this interpretation. The direction of the GARCH-$t$ vs. XGBoost-Pure comparison is consistent across all five tickers (Table 4), and the pooled analysis confirms the pattern with $p < 0.0001$. While pooling observations across correlated assets overstates the effective sample size and the pooled $p$-value should not be interpreted literally, the consistency of the sign across five independent estimation procedures provides stronger evidence than any single-ticker test. A risk manager who observes that one model class produces anti-conservative VaR estimates on every ticker tested has grounds to prefer the more conservative alternative, even when individual backtests are inconclusive.

## 7.4 A Decision Framework for Small Samples

We suggest the following decision protocol when the evaluation window contains fewer than 100 observations:

1. **Primary metric: directional conservatism.** Report $C = \hat{p} - \alpha$ for every candidate model. Models with $C < 0$ are conservative (fewer breaches than the nominal rate); models with $C > 0$ are anti-conservative. The magnitude of $C$ matters: at $n = 62$, a model with $C = +0.063$ breaches more than twice as often as the nominal rate.

2. **Cross-asset confirmation.** If the same directional pattern appears across multiple independent estimation exercises, the evidence is stronger than any single $p$-value. In our application, GARCH-$t$ produced $C < 0$ on all five tickers; XGBoost-Pure produced $C > 0$ on all five.

3. **Exact binomial $p$-values and Clopper–Pearson intervals.** Report these as supplementary diagnostics. The exact binomial test avoids the size distortion of the Kupiec test but remains underpowered. A non-significant $p$-value means the test is inconclusive, not that the model is adequate.

4. **Asymptotic tests for reference only.** Kupiec and Christoffersen $p$-values may be reported for comparability with the literature, but decisions should not rest on them at $n < 100$. A non-significant Kupiec $p$-value at $n = 62$ is expected under the null roughly 92% of the time, so it conveys little information about model quality.

Figure 6 summarises this decision protocol as a flowchart.

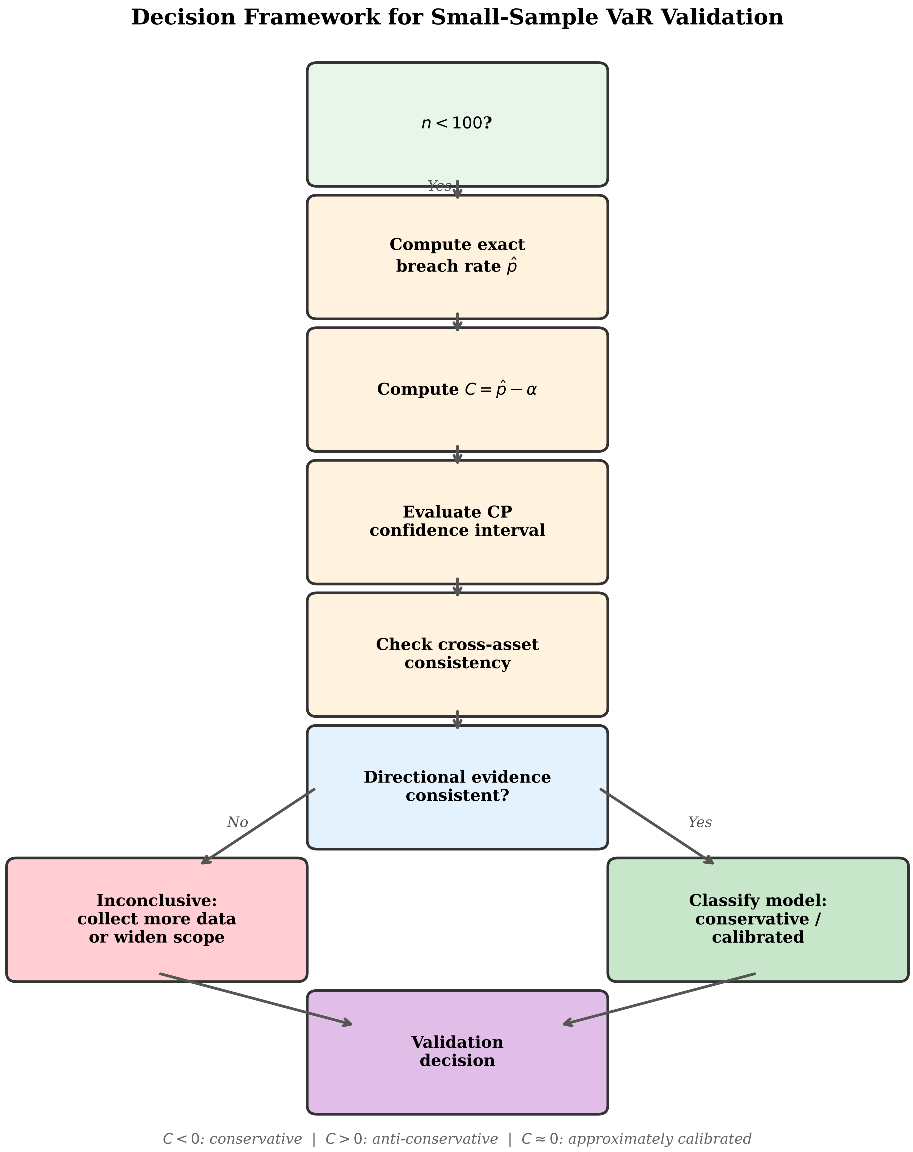


Figure 6: Decision framework for small-sample VaR validation.

## 7.5 Limitations

The numerical results depend on the specific configuration chosen here: a 125-day rolling estimation window, 20-day refit frequency, and a regime definition based on a 20-day volatility window with a 500-day lookback at the 90$^{\text{th}}$ percentile. Changing any of these parameters would shift the aligned sample size and potentially the breach-rate ranking. That said, the structural problem—that ex-ante regime definitions consume data and that standard backtests lose discriminatory power below 100 observations—is independent of the specific parameterization.

A Monte Carlo experiment using a GARCH(1,1)-$t$ data-generating process and 200 independent replications confirms that the Kupiec test rejects the true model 29.5% of the time at $n = 58$ (nominal 5%) and that no model achieves a Kupiec rejection rate exceeding 35% at that sample size (XGBoost-Q: 35.0%), establishing a ceiling on what backtests can achieve with short evaluation windows.

The XGBoost-$t$ variant uses a fixed $\nu = 5$ rather than estimating the degrees of freedom per refit window. Estimating $\nu$ from the data would be more consistent with the GARCH-$t$ specification but would also introduce additional parameter uncertainty in an already constrained setting. The pooled binomial test treats observations as independent despite the cross-sectional correlation among Mexican equity tickers; the reported pooled $p$-values are therefore indicative of direction and magnitude rather than exact. Future work with cluster-robust inference or panel methods would strengthen the pooled analysis. Finally, the model comparison here uses a single VaR level (95%) and does not assess Expected Shortfall performance; extending the framework to 99% VaR and ES would be a natural next step, though at $n = 62$ the expected number of 99% breaches is only 0.62, making any backtest essentially uninformative at that level. The decision framework proposed in Section 6.4 has not been validated on out-of-sample data or compared against alternative small-sample validation protocols; its recommendations reflect the diagnostic properties documented here, but its operational performance remains to be evaluated.

### 7.6 Relation to Basel Traffic-Light Approaches

Basel traffic-light frameworks Basel Committee on Banking Supervision (2019) assume sufficient observations to distinguish random variation from model misspecification. When effective sample sizes fall below 100 observations, traffic-light classifications become highly sensitive to single breaches and should therefore be interpreted cautiously.

### 7.7 Practical Implications for Model Governance

The findings suggest that model validation teams—operating under frameworks such as SR 11-7 Board of Governors of the Federal Reserve System (2011) with short histories, new products, or regime-conditioned samples—should supplement standard coverage tests with the diagnostics proposed here, since standard tests may fail to discriminate between acceptable and materially deficient models when sample sizes are small.

## 8 Conclusion

Standard VaR backtests become unreliable when the evaluation window falls below 100 observations, a condition that arises naturally under ex-ante volatility regime definitions. At $n = 62$, the Kupiec test has a Type I error of 7.6% (nominal 5%) and power below 50% against a model breaching at twice the nominal rate. The Christoffersen test is uninformative at these sample sizes. We propose augmenting traditional backtesting with exact binomial inference, Clopper–Pearson intervals, and directional conservatism metrics when sample sizes fall below 100 observations.

Applied to Mexican FIBRAs, this framework shows GARCH-$t$ producing breach rates at or below the nominal 5% level across all five tickers (pooled 3.6%), while XGBoost-based models consistently over-breach (pooled 9.7%; directional consistency across all five tickers, $p < 0.0001$ under independence). When $n < 100$, the question shifts from "which model fits best?" to "which model is least likely to produce dangerously anti-conservative VaR forecasts." Parsimony is not a preference in such settings—it is imposed by the data. Future work could extend this analysis to Expected Shortfall, individual securities, and alternative regime configurations.

# Declarations of Interest

The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the paper.

# Use of Artificial Intelligence

DeepSeek V4 Flash was used as an editing aid for sentence structure, clarity, and organisation of the manuscript. DeepSeek V4 Pro assisted with background research, literature verification, and formatting checks. All empirical results, figures, tables, and substantive claims are the product of independently written and executed Python scripts. The authors take full responsibility for the final content.

# Data and Code Availability

Source code will be available through a public repository upon publication. The pipeline is deterministic (global seed 42). Figures and tables are reproducible from cached data.

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